

AIDAN MURPHY

Senior Research Engineer

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PROFESSIONAL SUMMARY

I'm a research engineer with 6 years of experience building machine learning systems and distributed computing infrastructure. I've spent the last few years focused on deep learning optimization and cloud-native architectures, working mostly in Python and Go. I'm based in Dublin and enjoy solving complex problems that sit at the intersection of research and production systems.

TECHNICAL SKILLS

Machine Learning: TensorFlow, PyTorch, scikit-learn, Keras, LSTM, Transformer models, computer vision

Programming Languages: Python, Go, Java, C++, SQL, JavaScript

Cloud Platforms: AWS, Google Cloud Platform, Docker, Kubernetes

Data Engineering: Apache Spark, PostgreSQL, MongoDB, Redis, Kafka

Tools and Frameworks: Git, Jupyter, MLflow, DVC, FastAPI, Django

Research Areas: Neural network optimization, model compression, distributed training, federated learning

WORK EXPERIENCE

Senior Research Engineer

DataMind Labs

Dublin, Ireland

datamindlabs.io

January 2023 - Present

Led development of a federated learning framework that reduced model training time by 40% across 12 participating institutions while maintaining privacy guarantees

Designed and implemented a custom PyTorch training pipeline for vision transformers achieving state-of-the-art results on 5 benchmark datasets

Mentored 3 junior engineers on machine learning best practices and conducted weekly technical reviews of their research implementations

Published 2 peer-reviewed papers on model compression techniques and presented findings at international conferences

Research Engineer II

CloudScale Analytics

Dublin, Ireland

cloudscale.io

July 2020 - December 2022

Built distributed training infrastructure using Kubernetes that scaled TensorFlow models across 64 GPU nodes with <5% communication overhead

Developed Python package for automated hyperparameter optimization using Bayesian methods, reducing experimentation time from weeks to days

Implemented data pipeline processing 2TB daily of financial time series data using Apache Spark and Delta Lake

Collaborated with data science team to deploy 8 production models serving 15 million predictions monthly with 99.9% uptime

Machine Learning Engineer

TechVenture Solutions

Cork, Ireland

techventuresolutions.net

August 2018 - June 2020

Created end-to-end computer vision system for object detection using YOLOv3 and custom CNN architectures achieving 94% accuracy

Optimized model inference latency by 65% through quantization and pruning techniques, enabling deployment on edge devices

Developed REST API service in FastAPI handling 1000 requests per second across multiple AWS availability zones

Implemented continuous integration pipeline using GitHub Actions and automated retraining system triggered by data drift detection

EDUCATION

BSc Computer Science (Honours)

University College Dublin

Dublin, Ireland

Graduated: May 2018

Grade: 2.1

GITHUB REPOSITORIES

github.com/aidanmurphy/fedlearn-privacy - Federated learning framework with differential privacy guarantees

github.com/aidanmurphy/model-compression-toolkit - PyTorch utilities for knowledge distillation and quantization

github.com/aidanmurphy/distributed-training-benchmark - Benchmarking suite for distributed ML systems on Kubernetes

github.com/aidanmurphy/vision-transformer-optimization - Custom ViT implementations with various training strategies